

HOW THE CONFORMAL MODULUS OF AN ANNULUS QUANTITATIVELY CONTROLS HOLOMORPHIC MAPS

IZ TECÚM

ABSTRACT. Every nondegenerate doubly connected planar domain is conformally equivalent to $A(r, R)$ [3, 9], where

$$m := \text{mod}(A) = \frac{1}{2\pi} \log \frac{R}{r}.$$

We identify m with radial extremal length and reciprocal capacity [2, 15, 20]. We prove sharp centered Fourier–Laurent damping [7, 19], establish Schwarz–Pick estimates [6, 5, 12], and classify proper maps [9, 7]. We also describe boundary transfer and degeneration and quantify quasiconformal modulus distortion [17, 13, 22]. In round coordinates, a proper map of signed degree $n \in \mathbb{Z} \setminus \{0\}$ has

$$f(z) = c \cdot z^n, \quad c \in \mathbb{C} \setminus \{0\}, \quad \text{mod}(A') = |n| \cdot \text{mod}(A).$$

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Part 1. Modulus Control on Annuli

1. INTRODUCTION

Throughout, A is a nondegenerate doubly connected planar domain, $\text{mod}(A)$ is its conformal modulus, and every hyperbolic metric has curvature -1 ; the classical background is contained in [3, 7, 9]. Every such domain has a round model determined by one positive parameter.

$$A \cong A(r, R) := \{z \in \mathbb{C} : r < |z| < R\}, \quad m := \text{mod}(A) = \frac{1}{2\pi} \log\left(\frac{R}{r}\right),$$

$$q := \frac{r}{R} = e^{-2\pi m}, \quad A(r, R) \cong A(q, 1), \quad W := \log\left(\frac{R}{r}\right) = 2\pi m.$$

The logarithmic cover turns m into the Euclidean width W .

$$\begin{array}{ccc} S_W & \xrightarrow{w \mapsto e^w} & A(q, 1) \\ \downarrow w \sim w + 2\pi i & \nearrow \cong & \\ S_W / (2\pi i \mathbb{Z}) & & \end{array} \quad S_W := \{w \in \mathbb{C} : -W < \Re w < 0\}.$$

Variational geometry, periodic analysis, and covering equivariance consequently depend on the same parameter. Proper maps exhibit exact rigidity, whereas arbitrary holomorphic maps satisfy intrinsic contraction and mode-damping estimates rather than a monomial classification.

1.1. Result. The principal identities separate these geometric, analytic, and topological forms of control.

Theorem A. *Let $A = A(r, R)$, $m = \text{mod}(A)$, and $W = 2\pi m$. Then*

(1) *For the connecting and separating families Γ_{rad} and Γ_{sep} ,*

$$\text{Ext}_A(\Gamma_{\text{rad}}) = m, \quad \text{Ext}_A(\Gamma_{\text{sep}}) = m^{-1}, \quad \text{cap}(A) = m^{-1}.$$

(2) *For $C_* := \{|z| = \sqrt{rR}\}$,*

$$\rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log\left(\frac{|z|}{r}\right)\right), \quad \ell_A(C_*) = \frac{\pi}{m}.$$

(3) *If $f \in \text{Hol}(A, \mathbb{D})$ and $\rho_{\mathbb{D}}(\zeta) := 2(1 - |\zeta|^2)^{-1}$, then*

$$\rho_{\mathbb{D}}(f(z)) \cdot |f'(z)| \leq \rho_A(z), \quad d_{\mathbb{D}}(f(z), f(\zeta)) \leq d_A(z, \zeta).$$

(4) *If $A = A(q, 1)$, $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$, and $\sup_{A(q, 1)} |f| \leq M$, then*

$$|a_n| \leq M \quad (n \geq 0), \quad |a_{-n}| \leq M q^n \quad (n \geq 1), \quad \sup_{|z|=\sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1 - \sqrt{q}}.$$

(5) *Let $A' = A(r', R')$. If $g \in \text{Hol}(A, A')$ is proper of degree $n \in \mathbb{Z} \setminus \{0\}$, then, in round coordinates,*

$$g(z) = c \cdot z^n, \quad c \in \mathbb{C} \setminus \{0\}, \quad \text{mod}(A') = |n| \cdot \text{mod}(A).$$

(6) *If $h: A \rightarrow A'$ is a K -quasiconformal homeomorphism, then*

$$K^{-1} \cdot \text{mod}(A) \leq \text{mod}(A') \leq K \cdot \text{mod}(A).$$

Proof. § 9. □

2. ANNULI AND CONFORMAL MODULUS

The Riemann mapping theorem leaves no continuous conformal parameter for proper simply connected planar domains, whereas doubly connected domains retain one [1, 7, 9].

2.1. Logarithmic coordinates.

Definition 2.1.1 (Nondegenerate annulus, [3, 9]). A domain $A \subset \mathbb{C}$ is an *annulus* if $\widehat{\mathbb{C}} \setminus A$ has two connected components, and is *nondegenerate* if neither component is a point. For $0 < r < R$, write

$$A(r, R) := \{z \in \mathbb{C} : r < |z| < R\}.$$

Theorem 2.1.2 (Annulus uniformization, [3, 9, 15, 16]). *Every nondegenerate annulus is biholomorphic to $A(q, 1)$ for a unique $q \in (0, 1)$.*

Definition 2.1.3 (Conformal modulus, [1, 15, 22]). If $A \cong A(q, 1)$ as in Theorem 2.1.2, define

$$\text{mod}(A) := \frac{1}{2\pi} \log\left(\frac{1}{q}\right), \quad q = e^{-2\pi \text{mod}(A)}.$$

In particular,

$$A(r, R) \cong A(r/R, 1), \quad \text{mod}(A(r, R)) = \frac{1}{2\pi} \log\left(\frac{R}{r}\right).$$

Proposition 2.1.4 (Elementary normalizations, [1, 14]). *For $\lambda \in \mathbb{C} \setminus \{0\}$, scaling and inversion give*

$$z \mapsto \lambda z : A(r, R) \xrightarrow{\cong} A(|\lambda|r, |\lambda|R), \quad z \mapsto z^{-1} : A(r, R) \xrightarrow{\cong} A(R^{-1}, r^{-1}),$$

and both maps preserve modulus.

Proof. The radial inequalities determine both images.

$$r < |z| < R \iff |\lambda|r < |\lambda z| < |\lambda|R, \quad r < |z| < R \iff R^{-1} < |z^{-1}| < r^{-1}.$$

The maps have holomorphic inverses, and

$$\frac{|\lambda|R}{|\lambda|r} = \frac{1/r}{1/R} = \frac{R}{r} \xrightarrow{2.1.3} \text{mod}(A(|\lambda|r, |\lambda|R)) = \text{mod}(A(R^{-1}, r^{-1})) = \text{mod}(A(r, R)).$$

□

Corollary 2.1.5 (Complete conformal invariant, [3, 16]). *For nondegenerate annuli A and B ,*

$$A \cong B \iff \text{mod}(A) = \text{mod}(B).$$

Proof. Let $q_A, q_B \in (0, 1)$ be supplied by Theorem 2.1.2. Then

$$A \cong B \xLeftrightarrow{2.1.2} q_A = q_B \iff \log\left(\frac{1}{q_A}\right) = \log\left(\frac{1}{q_B}\right) \xLeftrightarrow{2.1.3} \text{mod}(A) = \text{mod}(B).$$

□

Fix $0 < r < R$, and put

$$S_{r,R} := \{w \in \mathbb{C} : \log r < \Re w < \log R\}, \quad W := \log\left(\frac{R}{r}\right).$$

Proposition 2.1.6 (Logarithmic cover, [3, 7, 9]). *The map $p(w) := e^w$ is a universal covering $p : S_{r,R} \rightarrow A(r, R)$, with*

$$\text{Deck}(p) = \{T_k : k \in \mathbb{Z}\}, \quad T_k(w) := w + 2\pi i k, \quad A(r, R) \cong S_{r,R}/(2\pi i \mathbb{Z}).$$

Its strip width is $W = 2\pi \text{mod}(A(r, R))$.

Proof. Write $w = u + iv$ and $z = \rho e^{i\theta}$. Then

$$w \in S_{r,R} \implies r < e^u = |e^w| < R, \quad z \in A(r,R) \implies \log \rho + i\theta \in S_{r,R}, \quad e^{\log \rho + i\theta} = z.$$

Moreover,

$$p'(w) = e^w \neq 0, \quad p(w_1) = p(w_2) \iff w_1 - w_2 \in 2\pi i\mathbb{Z}.$$

Each sufficiently small disc $U \Subset A(r,R)$ admits a logarithm branch L , and the disjoint sets $L(U) + 2\pi ik$ are the inverse sheets over U . Thus p is a covering; since $S_{r,R}$ is convex, it is universal, and the fiber identity gives $\text{Deck}(p) = \{T_k : k \in \mathbb{Z}\}$. Finally,

$$W = \log(R/r) \xrightarrow{2.1.3} W = 2\pi \text{mod}(A(r,R)).$$

□

Corollary 2.1.7 (Cylinder model, [1, 9]). *For*

$$\mathcal{R}_{r,R} := \{w \in \mathbb{C} : \log r < \Re w < \log R, 0 \leq \Im w \leq 2\pi\},$$

the horizontal sides are identified and

$$A(r,R) \cong \mathcal{R}_{r,R} / ((u,0) \sim (u,2\pi)), \quad \frac{W}{2\pi} = \text{mod}(A(r,R)).$$

Proof.

$$p(w + 2\pi i) = p(w) \xrightarrow{2.1.6} A(r,R) \cong \mathcal{R}_{r,R} / ((u,0) \sim (u,2\pi)).$$

$$W = 2\pi \text{mod}(A(r,R)) \xrightarrow{2.1.6} \frac{W}{2\pi} = \text{mod}(A(r,R)).$$

□

Figure 1 records the quotient.

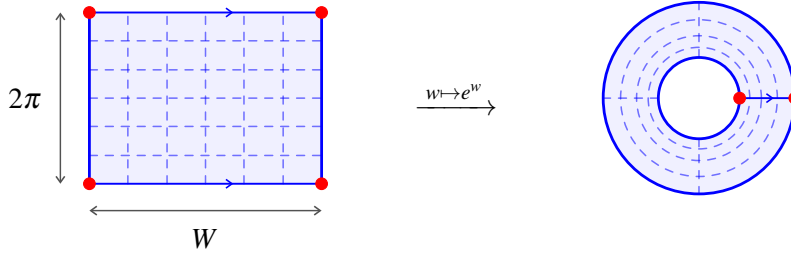


FIGURE 1. The exponential map identifies the horizontal edges, sends the vertical edges to the boundary circles, and carries the rectangular mesh to a logarithmic polar mesh.

Corollary 2.1.8 (Fundamental group, [7, 9]). *Every nondegenerate annulus A satisfies $\pi_1(A) \cong \mathbb{Z}$.*

Proof.

$$A \cong A(q,1) \xrightarrow{2.1.2} \pi_1(A) \cong \pi_1(A(q,1)) \xrightarrow{2.1.6} \text{Deck}(p) \cong \mathbb{Z}.$$

□

3. EXTREMAL LENGTH AND CAPACITY

The conformal modulus has two exact variational realizations: a length–area problem for transverse curve families and a Dirichlet problem for a condenser [2, 15, 20, 22].

3.1. Dual variational principles.

Definition 3.1.1 (Variational data). Let $\Omega \subset \mathbb{C}$, let Γ be a family of rectifiable curves in Ω , and let $\rho : \Omega \rightarrow [0, \infty]$ be Borel with $0 < \text{Area}_\rho(\Omega) < \infty$. Set

$$L_\rho(\Gamma) := \inf_{\gamma \in \Gamma} \int_\gamma \rho |dz|, \quad \text{Area}_\rho(\Omega) := \int_\Omega \rho^2 dx dy, \quad \text{Ext}_\Omega(\Gamma) := \sup_\rho \frac{L_\rho(\Gamma)^2}{\text{Area}_\rho(\Omega)}.$$

For a nondegenerate annulus A with boundary ends $\partial_0 A, \partial_1 A$, let $\mathcal{U}(A)$ be the Dirichlet class with traces 0 and 1 on those ends, and put

$$\mathcal{E}_A(u) := \int_A |\nabla u|^2 dx dy, \quad \text{cap}(A) := \inf_{u \in \mathcal{U}(A)} \mathcal{E}_A(u).$$

The traces are read on a round model furnished by 2.1.2; interchanging the ends replaces u by $1 - u$. We use the unnormalized capacity convention [15, 17, 21].

Proposition 3.1.2 (Conformal invariance). *Let $\phi : \Omega \rightarrow \Omega'$ be biholomorphic. Then*

$$\text{Ext}_\Omega(\Gamma) = \text{Ext}_{\Omega'}(\phi(\Gamma)), \quad \mathcal{E}_\Omega(v \circ \phi) = \mathcal{E}_{\Omega'}(v), \quad \text{cap}(\Omega) = \text{cap}(\Omega')$$

whenever the last expression concerns annuli [2, 15, 22].

Proof. Pull back an arbitrary density σ and potential v by $\rho = (\sigma \circ \phi)|\phi'|$ and $u = v \circ \phi$.

$$\int_\gamma \rho |dz| = \int_{\phi \circ \gamma} \sigma |dw|, \quad \text{Area}_\rho(\Omega) = \int_\Omega (\sigma^2 \circ \phi) |\phi'|^2 dx dy = \text{Area}_\sigma(\Omega').$$

Consequently,

$$\frac{L_\rho(\Gamma)^2}{\text{Area}_\rho(\Omega)} = \frac{L_\sigma(\phi(\Gamma))^2}{\text{Area}_\sigma(\Omega')} \stackrel{3.1.1}{\implies} \text{Ext}_\Omega(\Gamma) \geq \text{Ext}_{\Omega'}(\phi(\Gamma)),$$

while

$$|\nabla(v \circ \phi)|^2 = (|\nabla v|^2 \circ \phi) |\phi'|^2, \quad J_\phi = |\phi'|^2 \implies \mathcal{E}_\Omega(v \circ \phi) = \mathcal{E}_{\Omega'}(v).$$

Applying the same identities to ϕ^{-1} , and replacing v by $1 - v$ if ϕ exchanges the ends, proves all three claims. $\square$

Definition 3.1.3 (Dual curve families). For a nondegenerate annulus A , let $\Gamma_{\text{rad}}(A)$ be the rectifiable arcs joining its two boundary ends, with interiors in A , and let

$$\Gamma_{\text{sep}}(A) := \{\gamma : S^1 \rightarrow A : \gamma \text{ is rectifiable and } [\gamma] \neq 0 \text{ in } \pi_1(A)\}.$$

Thus $\Gamma_{\text{sep}}(A)$ is the essential, or separating, family; the identification $\pi_1(A) \cong \mathbb{Z}$ is recorded in 2.1.8 [2, 15, 20].

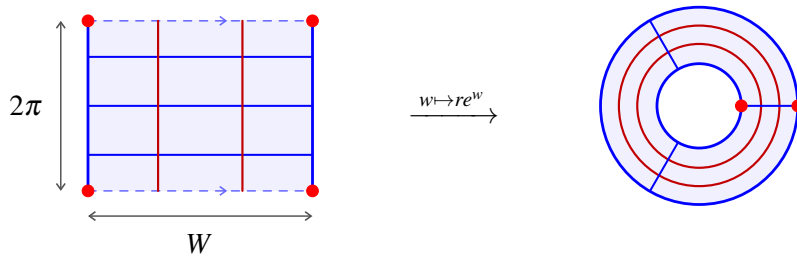


FIGURE 2. Under $w \mapsto re^w$, blue crosscuts project to radial curves and red vertical fibers to essential loops.

Theorem 3.1.4 (Variational characterization of modulus). *Let A be a nondegenerate annulus and set $m := \text{mod}(A)$. Then*

$$\text{Ext}_A(\Gamma_{\text{rad}}(A)) = m, \quad \text{Ext}_A(\Gamma_{\text{sep}}(A)) = \text{cap}(A) = m^{-1}.$$

In particular,

$$\text{cap}(A) = \text{Ext}_A(\Gamma_{\text{sep}}(A)) = \text{Ext}_A(\Gamma_{\text{rad}}(A))^{-1}, \quad \text{Ext}_A(\Gamma_{\text{rad}}(A))\text{Ext}_A(\Gamma_{\text{sep}}(A)) = 1.$$

If $A = A(r, R)$ and $W := \log(R/r)$, then $\rho_(z) := |z|^{-1}$ is extremal for both curve families, and the unique capacity minimizer is*

$$u_*(z) := \frac{\log(|z|/r)}{W}.$$

These normalizations are classical [2, 15, 17, 20, 22].

Proof. Uniformization and conformal invariance reduce all three problems to one Euclidean cylinder.

$$C_W := (0, W) \times (\mathbb{R}/(2\pi\mathbb{Z})) \xleftrightarrow{2.1.7} A(r, R), \quad W = \log(R/r) = 2\pi m \xleftrightarrow{2.1.4} m = \frac{W}{2\pi}.$$

For an admissible density $\tilde{\rho}$ on C_W , the horizontal segments and vertical fibers in 2 give

$$\begin{aligned} 2\pi L_{\tilde{\rho}}(\tilde{\Gamma}_{\text{rad}}) &\leq \int_0^{2\pi} \int_0^W \tilde{\rho} dx dy \leq (2\pi W)^{1/2} \text{Area}_{\tilde{\rho}}(C_W)^{1/2} \implies \frac{L_{\tilde{\rho}}(\tilde{\Gamma}_{\text{rad}})^2}{\text{Area}_{\tilde{\rho}}(C_W)} \leq \frac{W}{2\pi}, \\ WL_{\tilde{\rho}}(\tilde{\Gamma}_{\text{sep}}) &\leq \int_0^W \int_0^{2\pi} \tilde{\rho} dy dx \leq (2\pi W)^{1/2} \text{Area}_{\tilde{\rho}}(C_W)^{1/2} \implies \frac{L_{\tilde{\rho}}(\tilde{\Gamma}_{\text{sep}})^2}{\text{Area}_{\tilde{\rho}}(C_W)} \leq \frac{2\pi}{W}. \end{aligned}$$

For $\tilde{\rho}_* \equiv 1$, connecting curves have length at least W , while every essential loop lifts with endpoint displacement $2\pi ik$, $k \in \mathbb{Z} \setminus \{0\}$, and hence has length at least 2π ; horizontal and vertical fibers attain both bounds. Therefore

$$L_{\tilde{\rho}_*}(\tilde{\Gamma}_{\text{rad}}) = W, \quad L_{\tilde{\rho}_*}(\tilde{\Gamma}_{\text{sep}}) = 2\pi, \quad \text{Area}_{\tilde{\rho}_*}(C_W) = 2\pi W \implies \tilde{\rho}_* \equiv 1 \xrightarrow{3.1.1} \text{Ext}_{C_W}(\tilde{\Gamma}_{\text{rad}}) = \frac{W}{2\pi}, \quad \text{Ext}_{C_W}(\tilde{\Gamma}_{\text{sep}}) = \frac{2\pi}{W}.$$

For $\tilde{u} \in \mathcal{U}(C_W)$, the trace values and Cauchy–Schwarz yield, for almost every y ,

$$1 = \int_0^W \partial_x \tilde{u}(x, y) dx \leq W^{1/2} \left(\int_0^W |\partial_x \tilde{u}(x, y)|^2 dx \right)^{1/2} \implies \mathcal{E}_{C_W}(\tilde{u}) \geq \int_0^{2\pi} \int_0^W |\partial_x \tilde{u}|^2 dx dy \geq \frac{2\pi}{W}.$$

The affine potential attains this bound, and its equality conditions give uniqueness:

$$\begin{aligned} \tilde{u}_*(x, y) &:= \frac{x}{W}, \quad \mathcal{E}_{C_W}(\tilde{u}_*) = \int_0^{2\pi} \int_0^W \frac{1}{W^2} dx dy = \frac{2\pi}{W}. \\ \mathcal{E}_{C_W}(\tilde{u}) = \frac{2\pi}{W} &\implies \partial_y \tilde{u} = 0, \quad \partial_x \tilde{u} = W^{-1} \implies \tilde{u} = \tilde{u}_*. \end{aligned}$$

Transporting these extremizers through the logarithmic model gives

$$\begin{aligned} \tilde{\rho}_* \equiv 1 &\xleftrightarrow{2.1.6} \rho_*(z) = \frac{1}{|z|}, \quad \tilde{u}_*(x, y) = \frac{x}{W} \xleftrightarrow{2.1.7} u_*(z) = \frac{\log(|z|/r)}{W} \implies \frac{W}{2\pi} = m, \\ \frac{2\pi}{W} &= m^{-1} \xrightarrow{2.1.2, 3.1.2} \text{Ext}_A(\Gamma_{\text{rad}}) = m, \quad \text{Ext}_A(\Gamma_{\text{sep}}) = \text{cap}(A) = m^{-1}. \end{aligned}$$

This proves the theorem. $\square$

4. COVERINGS AND FOURIER ANALYSIS

Fix $A := A(r, R)$ and put

$$s := \sqrt{rR}, \quad W := \log(R/r) \xrightarrow{2.1.4} W = 2\pi \bmod(A), \quad S_W := \{w \in \mathbb{C} : |\Re w| < W/2\}.$$

By Proposition 2.1.6, $p_s(w) := se^w$ has deck generator $T(w) := w + 2\pi i$; pullback by p_s converts Laurent monomials into periodic Fourier modes [9, 7, 19].

4.1. Mode damping.

Proposition 4.1.1 (Periodic descent, [9, 7]). *For the centered cover $p_s : S_W \rightarrow A$, composition with p_s gives*

$$\mathrm{Hol}(A, \mathbb{C}) \xrightarrow{\cong} \{F \in \mathrm{Hol}(S_W, \mathbb{C}) : F \circ T = F\}, \quad f \mapsto F := f \circ p_s.$$

Equivalently, the following diagram commutes, and the dashed arrow is unique.

$$\begin{array}{ccccc} & & F & & \\ & \nearrow T & & \searrow F & \\ S_W & \xrightarrow{\quad} & S_W & \xrightarrow{\quad} & \mathbb{C} \\ p_s \downarrow & & \downarrow p_s & \nearrow f & \\ A & \xlongequal{\quad} & A & & \end{array}$$

Moreover,

$$F'(w) = se^w f'(se^w), \quad z = p_s(w) \implies z f'(z) = F'(w).$$

Proof. The fiber relation from Proposition 2.1.6 governs the descent.

$$p_s \circ T = p_s, \quad F = f \circ p_s \implies F \circ T = f \circ p_s \circ T = f \circ p_s = F.$$

$$p_s(w_1) = p_s(w_2) \xrightarrow{2.1.6} w_2 - w_1 \in 2\pi i\mathbb{Z}, \quad F \circ T = F \implies F(w_2) = F(w_1).$$

$$f(p_s(w)) := F(w), \quad p_s|_U : U \xrightarrow{\cong} V \implies f|_V = F \circ (p_s|_U)^{-1} \in \mathrm{Hol}(V, \mathbb{C}).$$

$$f_1 \circ p_s = f_2 \circ p_s, \quad p_s(S_W) = A \implies f_1 = f_2, \quad F = f \circ p_s \implies F'(w) = se^w f'(se^w).$$

This proves the descent bijection and derivative identity. $\square$

Theorem 4.1.2 (Fourier–Laurent correspondence, [7, 18, 19]). *Let $f \in \mathrm{Hol}(A, \mathbb{C})$ and let $F := f \circ p_s$. There are unique coefficients $(a_n)_{n \in \mathbb{Z}}$ and $(b_n)_{n \in \mathbb{Z}}$ such that*

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n, \quad F(w) = \sum_{n \in \mathbb{Z}} b_n e^{nw}, \quad b_n = s^n a_n,$$

with absolute and locally uniform convergence. For $r < \rho < R$, $|x| < W/2$, and $n \in \mathbb{Z}$,

$$a_n = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}} dz, \quad b_n = \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny} dy.$$

Proof. Cauchy's theorem and Laurent's theorem identify the two expansions.

$$a_n(\rho) := \frac{1}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}} dz, \quad a_n(\rho_2) - a_n(\rho_1) = \frac{1}{2\pi i} \int_{\partial A(\rho_1, \rho_2)} \frac{f(z)}{z^{n+1}} dz = 0.$$

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n \implies F(w) = f(se^w) = \sum_{n \in \mathbb{Z}} (s^n a_n) e^{nw} \implies b_n = s^n a_n.$$

$$z = se^{x+iy}, \quad dz = ise^{x+iy} dy \implies b_n = s^n a_n = \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny} dy.$$

$$K \subseteq S_W, \quad p_s(K) \subseteq A \implies \sum_{n \in \mathbb{Z}} a_n (se^w)^n \longrightarrow f(se^w) = F(w) \quad \text{uniformly on } K.$$

This proves the Fourier–Laurent dictionary. $\square$

Theorem 4.1.3 (Sharp mode damping, [7, 8]). *Let $f \in \text{Hol}(A, \mathbb{C})$ satisfy $\|f\|_{L^\infty(A)} \leq M$, and let b_n be the centered coefficients in Theorem 4.1.2. Every mode satisfies*

$$|b_n| \leq M e^{-|n|W/2} = M e^{-\pi|n|\text{mod}(A)} \quad (n \in \mathbb{Z}),$$

and the exponent and constant are sharp for each $n \neq 0$.

Proof. The coefficient line may be moved to the favorable side of the strip.

$$|b_n| \xrightarrow{4.1.2} \frac{e^{-nx}}{2\pi} \left| \int_0^{2\pi} F(x+iy) e^{-iny} dy \right| \leq \frac{e^{-nx}}{2\pi} \int_0^{2\pi} |F(x+iy)| dy \leq M e^{-nx}.$$

$$\begin{aligned} n \geq 0, \quad x \uparrow W/2 &\implies |b_n| \leq M e^{-nW/2}, & n < 0, \quad x \downarrow -W/2 &\implies |b_n| \leq M e^{-|n|W/2}. \\ W = 2\pi \text{mod}(A) &\xrightarrow{2.1.4} e^{-|n|W/2} = e^{-\pi|n|\text{mod}(A)}, & \frac{s}{R} = \frac{r}{s} = e^{-W/2}. \end{aligned}$$

$$f_k^+(z) := M \left(\frac{z}{R} \right)^k, \quad f_k^-(z) := M \left(\frac{r}{z} \right)^k, \quad \|f_k^\pm\|_{L^\infty(A)} = M, \quad b_k^+ = b_{-k}^- = M e^{-kW/2} \quad (k \geq 1).$$

This proves the sharp damping estimate. $\square$

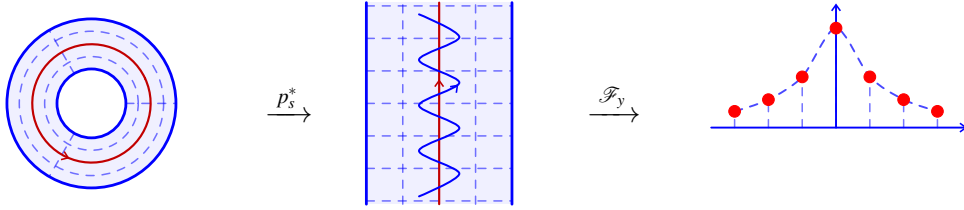


FIGURE 3. Pullback by $p_s(w) = se^w$ sends $a_n z^n$ to $b_n e^{nw}$, where $b_n = s^n a_n$, and the centered strip yields $|b_n| \leq M e^{-|n|W/2}$.

Corollary 4.1.4 (Core-circle rigidity). *Let $m := \text{mod}(A)$, $\delta := e^{-\pi m}$, and $\|f\|_{L^\infty(A)} \leq M$. On the core circle $|z| = s$ in Figure 3,*

$$\sup_{|z|=s} |f(z) - a_0| \leq \frac{2M\delta}{1-\delta}, \quad \sup_{|z|=s} |zf'(z)| \leq \frac{2M\delta}{(1-\delta)^2}.$$

For $A = A(q, 1)$, Proposition 2.1.4 gives

$$s = \sqrt{q}, \quad \delta = \sqrt{q}, \quad \sup_{|z|=\sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1-\sqrt{q}}, \quad \sup_{|z|=\sqrt{q}} |zf'(z)| \leq \frac{2M\sqrt{q}}{(1-\sqrt{q})^2}.$$

Proof. The centered Fourier expansion and Theorem 4.1.3 control both tails.

$$z = se^{iy} \implies f(z) - a_0 = \sum_{n \neq 0} b_n e^{iny}, \quad zf'(z) = \sum_{n \neq 0} n b_n e^{iny}.$$

$$\sup_{|z|=s} |f(z) - a_0| \leq \sum_{n \neq 0} |b_n| \xrightarrow{4.1.3} M \sum_{n \neq 0} \delta^{|n|} = 2M \sum_{n \geq 1} \delta^n = \frac{2M\delta}{1-\delta}.$$

$$\sup_{|z|=s} |zf'(z)| \leq \sum_{n \neq 0} |n| |b_n| \xrightarrow{4.1.3} 2M \sum_{n \geq 1} n \delta^n = \frac{2M\delta}{(1-\delta)^2}.$$

This proves exponential rigidity on the core circle. $\square$

Thus the cylinder length $W = 2\pi \text{mod}(A)$ is the exact spectral separation scale, and every nonconstant core mode loses the factor $e^{-\pi|n|\text{mod}(A)}$ [7, 8].

5. HYPERBOLIC GEOMETRY AND SCHWARZ–PICK

The centered cover from Proposition 2.1.6 turns the complete curvature- -1 metric into an explicit transverse density. Its core geometry and Schwarz–Pick contraction make the dependence on $\text{mod}(A)$ intrinsic [6, 5, 12].

5.1. Intrinsic estimates.

Definition 5.1.1 (Hyperbolic metric, [6, 5]). Let $\Omega \subset \mathbb{C}$ have universal cover $\mathbb{D}$, and let $p : \mathbb{D} \rightarrow \Omega$ be a covering. Its Poincaré density, curve length, and distance are normalized by

$$\rho_{\mathbb{D}}(\zeta) := \frac{2}{1-|\zeta|^2}, \quad \rho_{\mathbb{H}}(\zeta) := \frac{1}{\Im(\zeta)}, \quad \rho_{\Omega}(p(\zeta))|p'(\zeta)| = \rho_{\mathbb{D}}(\zeta), \quad ds_{\Omega} := \rho_{\Omega}(z)|dz|.$$

$$\ell_{\Omega}(\gamma) := \int_{\gamma} \rho_{\Omega}(z)|dz|, \quad d_{\Omega}(z_0, z_1) := \inf_{\gamma: z_0 \rightsquigarrow z_1} \ell_{\Omega}(\gamma).$$

Theorem 5.1.2 (Explicit annular density, [6, 5, 12]). Let $A = A(r, R)$, $s := \sqrt{rR}$, $W := \log(R/r)$, and $m := \text{mod}(A)$. Then

$$W = 2\pi m, \quad \rho_A(z) = \frac{\pi}{W|z|} \sec\left(\frac{\pi}{W} \log \frac{|z|}{s}\right) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log \frac{|z|}{r}\right), \quad \rho_A(se^{i\theta}) = \frac{1}{2ms}.$$

Proof. We pull back the half-plane metric through the centered cover.

$$S_W := \{w \in \mathbb{C} : |\Re(w)| < W/2\}, \quad \Phi(w) := i \exp(\pi i w / W), \quad \Phi : S_W \xrightarrow{\sim} \mathbb{H}.$$

$$x := \Re(w), \quad \Im(\Phi(w)) = |\Phi(w)| \cos(\pi x / W), \quad |\Phi'(w)| = \frac{\pi}{W} |\Phi(w)| \implies \rho_{S_W}(w) = \frac{\pi}{W} \sec(\pi x / W).$$

$$p_s(w) := se^w, \quad |p'_s(w)| = |p_s(w)| \xrightarrow{2.1.6, 5.1.1} \rho_A(p_s(w)) |p_s(w)| = \rho_{S_W}(w).$$

$$x = \log \frac{|z|}{s}, \quad \log \frac{|z|}{r} = x + \frac{W}{2}, \quad \rho_A(z) = \frac{\pi}{W|z|} \sec\left(\frac{\pi x}{W}\right) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log \frac{|z|}{r}\right).$$

$$m = \text{mod}(A) \xleftrightarrow{2.1.4} W = 2\pi m, \quad |z| = s \implies \rho_A(se^{i\theta}) = \frac{\pi}{Ws} = \frac{1}{2ms}.$$

This proves the density formula and its core value. $\square$

Proposition 5.1.3 (Intrinsic geometry, [11, 5, 12]). Let $A = A(r, R)$, $s := \sqrt{rR}$, $m := \text{mod}(A)$, $W := \log(R/r) = 2\pi m$, and

$$\Delta_A(z) := \min \left\{ \log \frac{|z|}{r}, \log \frac{R}{|z|} \right\} = \frac{W}{2} - \left| \log \frac{|z|}{s} \right|.$$

The density satisfies

$$\frac{1}{|z| \Delta_A(z)} \leq \rho_A(z) \leq \frac{\pi}{2|z| \Delta_A(z)}.$$

For $r < \rho_1, \rho_2 < R$, $t_j := \log(\rho_j/r)$, and $\theta \in \mathbb{R}$,

$$d_A(\rho_1 e^{i\theta}, \rho_2 e^{i\theta}) = \left| \log \frac{\tan(\pi t_2 / (2W))}{\tan(\pi t_1 / (2W))} \right|.$$

Let $\Gamma_1(A)$ be the rectifiable closed curves representing a generator of $\pi_1(A)$. Then

$$C_* := \{|z| = s\} \text{ is a hyperbolic geodesic,} \quad \ell_A(C_*) = \frac{2\pi^2}{W} = \frac{\pi}{m}.$$

$\min_{\gamma \in \Gamma_1(A)} \ell_A(\gamma) = \ell_A(C_*)$, $\ell_A(\gamma) = \ell_A(C_*) \iff \gamma \text{ parametrizes } C_* \text{ once without backtracking.}$

Proof. Write every generator lift as $\tilde{\gamma} = X + iY$ and use the models in Figure 4.

$$\begin{aligned}
 u &:= \frac{\pi \Delta_A(z)}{W} \in (0, \pi/2] \xrightarrow{5.1.2} \rho_A(z) |z| \Delta_A(z) = \frac{u}{\sin u}, \quad \frac{2u}{\pi} \leq \sin u \leq u \implies 1 \leq \frac{u}{\sin u} \leq \frac{\pi}{2}. \\
 \ell_A(\gamma) &\geq \left| \int_{t_1}^{t_2} \frac{\pi}{W} \csc(\pi t/W) dt \right| = \left| \log \frac{\tan(\pi t_2/(2W))}{\tan(\pi t_1/(2W))} \right|, \quad \gamma(\tau) = e^{(1-\tau) \log \rho_1 + \tau \log \rho_2 + i\theta} \implies \text{equality}. \\
 \Phi(i\mathbb{R}) = i\mathbb{R}_{>0} &\text{ is geodesic in } \mathbb{H} \xrightarrow{5.1.2} C_* = p_s(i\mathbb{R}) \text{ is geodesic,} \quad \ell_A(C_*) = 2\pi \cdot \frac{\pi}{W} = \frac{2\pi^2}{W} = \frac{\pi}{m}. \\
 Y(1) - Y(0) &= \pm 2\pi, \quad \rho_{S_W}(X + iY) \geq \frac{\pi}{W}, \quad |\tilde{\gamma}| \geq |Y'| \implies \ell_A(\gamma) \geq \frac{\pi}{W} \int_0^1 |Y'(t)| dt \geq \frac{2\pi^2}{W}. \\
 \ell_A(\gamma) &= \frac{2\pi^2}{W} \iff X \equiv 0 \wedge Y \text{ is monotone} \iff \gamma \text{ parametrizes } C_* \text{ once without backtracking.}
 \end{aligned}$$

This proves the comparison, distance, and core-minimality claims. $\square$

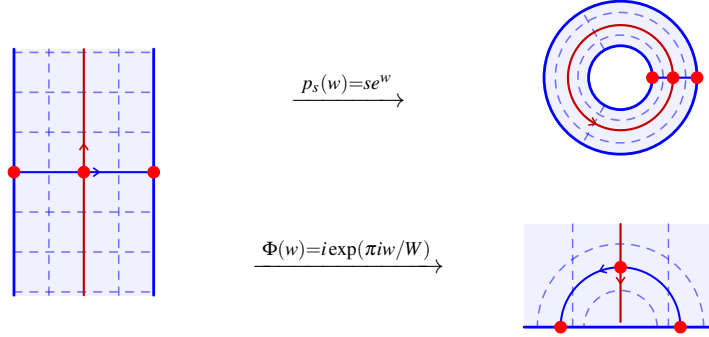


FIGURE 4. The centered strip sends its red midline to the annular core under p_s and to $i\mathbb{R}_{>0}$ under Φ ; its blue transverse geodesic becomes a radius and a semicircle.

Theorem 5.1.4 (Schwarz–Pick, [1, 7, 5, 6]). *Let $\Omega, \Omega' \subset \mathbb{C}$ be hyperbolic domains, and let $f \in \text{Hol}(\Omega, \Omega')$. Then*

$$\rho_{\Omega'}(f(z)) |f'(z)| \leq \rho_{\Omega}(z), \quad d_{\Omega'}(f(z_0), f(z_1)) \leq d_{\Omega}(z_0, z_1).$$

Proof. Fix $z \in \Omega$ and lift f through normalized disk covers.

$$\begin{array}{ccc}
 (\mathbb{D}, 0) & \xrightarrow{\tilde{f}} & (\mathbb{D}, 0) \\
 p \downarrow & & \downarrow q \\
 (\Omega, z) & \xrightarrow{f} & (\Omega', f(z))
 \end{array} \quad q \circ \tilde{f} = f \circ p, \quad \tilde{f}(0) = 0.$$

$$|\tilde{f}'(0)| \leq 1, \quad q'(0) \tilde{f}'(0) = f'(z) p'(0) \implies |f'(z)| |p'(0)| \leq |q'(0)|.$$

$$p, q \text{ universal covers} \xrightarrow{5.1.1} \rho_{\Omega}(z) |p'(0)| = \rho_{\mathbb{D}}(0) = 2 = \rho_{\Omega'}(f(z)) |q'(0)| \implies \rho_{\Omega'}(f(z)) |f'(z)| \leq \rho_{\Omega}(z).$$

$$\ell_{\Omega'}(f \circ \gamma) = \int_{\gamma} \rho_{\Omega'}(f(z)) |f'(z)| |dz| \leq \int_{\gamma} \rho_{\Omega}(z) |dz| = \ell_{\Omega}(\gamma) \implies d_{\Omega'}(f(z_0), f(z_1)) \leq d_{\Omega}(z_0, z_1).$$

This proves the infinitesimal and global contractions. $\square$

Corollary 5.1.5 (Annulus-to-disk estimates). *Let $A = A(r, R)$, $s := \sqrt{rR}$, $m := \text{mod}(A)$, $W := \log(R/r) = 2\pi m$, and $f \in \text{Hol}(A, \mathbb{D})$. For*

$$\vartheta(\theta, \varphi) := \min_{k \in \mathbb{Z}} |\theta - \varphi + 2\pi k|,$$

one has

$$\begin{aligned} \frac{|zf'(z)|}{1-|f(z)|^2} &\leq \frac{\pi}{2W} \sec\left(\frac{\pi}{W} \log \frac{|z|}{s}\right), & |f'(z)| &\leq \frac{\pi(1-|f(z)|^2)}{4|z|\Delta_A(z)} \leq \frac{\pi}{4|z|\Delta_A(z)}. \\ \frac{|se^{i\theta} f'(se^{i\theta})|}{1-|f(se^{i\theta})|^2} &\leq \frac{1}{4m}, & d_{\mathbb{D}}(f(se^{i\theta}), f(se^{i\varphi})) &\leq \frac{\vartheta(\theta, \varphi)}{2m}. \end{aligned}$$

Proof. We insert the explicit density into Schwarz–Pick.

$$\begin{aligned} f \in \text{Hol}(A, \mathbb{D}) &\xrightarrow{5.1.4} \frac{2|f'(z)|}{1-|f(z)|^2} \leq \rho_A(z) \xrightarrow{5.1.2} \frac{|zf'(z)|}{1-|f(z)|^2} \leq \frac{\pi}{2W} \sec\left(\frac{\pi}{W} \log \frac{|z|}{s}\right). \\ \frac{2|f'(z)|}{1-|f(z)|^2} \leq \rho_A(z) &\xrightarrow{5.1.3} \frac{2|f'(z)|}{1-|f(z)|^2} \leq \frac{\pi}{2|z|\Delta_A(z)} \implies |f'(z)| \leq \frac{\pi(1-|f(z)|^2)}{4|z|\Delta_A(z)}. \\ z = se^{i\theta}, \quad W = 2\pi m &\implies \frac{|zf'(z)|}{1-|f(z)|^2} \leq \frac{\pi}{2W} = \frac{1}{4m}. \\ \gamma_{\theta, \varphi} \subset C_* \text{ is the shorter arc,} &\quad \ell_A(\gamma_{\theta, \varphi}) = \frac{\pi}{W} \vartheta(\theta, \varphi) = \frac{\vartheta(\theta, \varphi)}{2m}. \\ f \in \text{Hol}(A, \mathbb{D}) &\xrightarrow{5.1.4} d_{\mathbb{D}}(f(se^{i\theta}), f(se^{i\varphi})) \leq d_A(se^{i\theta}, se^{i\varphi}) \leq \ell_A(\gamma_{\theta, \varphi}) = \frac{\vartheta(\theta, \varphi)}{2m}. \end{aligned}$$

This proves the derivative and angular estimates. $\square$

Together with Theorem 4.1.3, these estimates separate the intrinsic scale m^{-1} from the Fourier scale $e^{-\pi m}$ [6, 12].

6. HOLOMORPHIC MAPS BETWEEN ANNULI

Throughout, $A := A(r, R)$, $A' := A(r', R')$, $s := \sqrt{rR}$, $s' := \sqrt{r'R'}$, $W := \log(R/r)$, and $W' := \log(R'/r')$, with centered covers $p_s : S_W \rightarrow A$, $p_{s'} : S_{W'} \rightarrow A'$, and deck generator $T(w) := w + 2\pi i$; properness forces exact modulus scaling [3, 7, 9].

6.1. Degree and rigidity.

Definition 6.1.1 (Properness and degree, [7, 9]). For a continuous map $f : A \rightarrow A'$, properness and degree under the counterclockwise identifications in Corollary 2.1.8 are specified by

$$f \text{ proper} \iff f^{-1}(K) \Subset A \quad \text{for every } K \Subset A', \quad f_* : \mathbb{Z} \longrightarrow \mathbb{Z}, \quad k \longmapsto \deg(f) \cdot k.$$

For $f \in \text{Hol}(A, A')$ and a piecewise C^1 counterclockwise generator γ ,

$$\deg(f) = \text{wind}_0(f \circ \gamma) = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz.$$

Proposition 6.1.2 (Equivariant lift, [7, 9]). *Every $f \in \text{Hol}(A, A')$ admits a holomorphic lift $F : S_W \rightarrow S_{W'}$ and a unique $n \in \mathbb{Z}$ such that*

$$\begin{array}{ccc} S_W & \xrightarrow{F} & S_{W'} \\ p_s \downarrow & & \downarrow p_{s'} \\ A & \xrightarrow{f} & A' \end{array} \quad F \circ T = T^n \circ F, \quad n = \deg(f).$$

Moreover,

$$|\deg(f)| \cdot \text{mod}(A) \leq \text{mod}(A'), \quad \deg(f) = 0 \iff f = e^g \text{ for some } g \in \text{Hol}(A, \mathbb{C}).$$

Proof. Lift $f \circ p_s$ through the logarithmic cover of A' and average its real part across one deck period.

$$\begin{aligned} p_{s'} \circ F &= f \circ p_s, & p_s \circ T &= p_s \implies F(T(w)) - F(w) \in 2\pi i\mathbb{Z}. \\ \frac{F(T(w)) - F(w)}{2\pi i} &\in \text{Hol}(S_W, \mathbb{Z}), & S_W \text{ connected} &\implies F \circ T = T^n \circ F \xleftrightarrow{6.1.1} n = \deg(f). \\ M(x) &:= \frac{1}{2\pi} \int_0^{2\pi} \Re(F(x + iy)) dy, & M'(x) &= \frac{\Im(F(x + 2\pi i)) - \Im(F(x))}{2\pi} = n, \\ |\Re(F(w))| &< \frac{W'}{2}, & M(x) = n \cdot x + a &\implies |n| \cdot W \leq W' \xleftrightarrow{2.1.4} |n| \cdot \text{mod}(A) \leq \text{mod}(A'). \\ n = 0 &\iff F \circ T = F \xleftrightarrow{4.1.1} F = G \circ p_s \iff f = e^g, & g &:= \log(s') + G. \end{aligned}$$

This proves the lift, degree, modulus, and logarithm claims. $\square$

Theorem 6.1.3 (Proper-map classification, [3, 7, 9]). *Let $f \in \text{Hol}(A, A')$ be proper and put $n := \deg(f)$. Then*

$$\begin{aligned} n &\neq 0, & f(z) &= c \cdot z^n \quad (c \in \mathbb{C} \setminus \{0\}), & \text{mod}(A') &= |n| \cdot \text{mod}(A), \\ \{|c|^{r'}, |c|R^n\} &= \{r', R'\}, & f : A &\rightarrow A' \text{ is an } |n|\text{-sheeted unbranched covering.} \end{aligned}$$

Proof. Let F be the lift from Proposition 6.1.2; properness excludes constants, while openness, closedness, and connected collars outside compact middle preimages give the uniform end limits below.

$$\begin{aligned} \Re(F(x + iy)) &\longrightarrow \varepsilon_{\pm} \frac{W'}{2} \quad (x \rightarrow \pm W/2), & \varepsilon_+ &= -\varepsilon_- =: \varepsilon \in \{1, -1\}. \\ h(z) &:= \log \frac{|f(z)|}{s'} - \varepsilon \frac{W'}{W} \log \frac{|z|}{s}, & \Delta h &= 0, & h(z) \rightarrow 0 \quad (z \rightarrow \partial A) &\implies h = 0. \\ h = 0 &\implies \Re(F(w)) = \varepsilon \frac{W'}{W} \Re(w), & \Re\left(F(w) - \varepsilon \frac{W'}{W} w\right) &= 0 \implies F(w) = \varepsilon \frac{W'}{W} w + i\theta \quad (\theta \in \mathbb{R}). \\ F \circ T &= T^n \circ F \xrightarrow{6.1.2} 2\pi i n = F(T(w)) - F(w) = 2\pi i \varepsilon \frac{W'}{W} \implies n = \varepsilon \frac{W'}{W} \neq 0, & W' &= |n| \cdot W. \\ z = p_s(w) &\implies f(z) = p_{s'}(F(w)) = s' e^{i\theta} \left(\frac{z}{s}\right)^n = c \cdot z^n, & c &:= s' e^{i\theta} s^{-n} \neq 0, & f'(z) &= n \cdot c \cdot z^{n-1} \neq 0. \end{aligned}$$

This proves the classification, covering, and modulus law. $\square$

Corollary 6.1.4 (Existence and self-maps). *Let X, Y be nondegenerate annuli and $n \in \mathbb{Z} \setminus \{0\}$. Then*

$$\exists f \in \text{Hol}(X, Y) \text{ proper with } \deg(f) = n \iff \text{mod}(Y) = |n| \cdot \text{mod}(X).$$

Every proper holomorphic self-map of X is biholomorphic; for a round annulus,

$$f : A(r, R) \rightarrow A(r, R) \text{ proper} \iff f(z) = e^{i\theta} z \text{ or } f(z) = e^{i\theta} \frac{rR}{z} \quad (\theta \in \mathbb{R}).$$

Proof. Choose degree-one uniformizations $\phi : X \rightarrow A(q, 1)$ and $\psi : Y \rightarrow A(q', 1)$ by Theorem 2.1.2, postcomposing with $z \mapsto q/z$ and $z \mapsto q'/z$ if necessary.

$$\begin{aligned} \tilde{f} &:= \psi \circ f \circ \phi^{-1}, & \deg(\tilde{f}) &= n, & f \text{ proper} &\iff \tilde{f} \text{ proper} \xrightarrow{6.1.3} \text{mod}(Y) = |n| \cdot \text{mod}(X). \\ & & \text{mod}(Y) &= |n| \cdot \text{mod}(X) \xleftrightarrow{2.1.4} q' = q^{|n|}. \\ \mu_n(z) &:= \begin{cases} z^n, & n > 0, \\ q^{|n|} \cdot z^n, & n < 0, \end{cases} & \mu_n : A(q, 1) &\rightarrow A(q', 1) \text{ proper}, & \deg(\mu_n) &= n. \\ f_n &:= \psi^{-1} \circ \mu_n \circ \phi : X \rightarrow Y, & f_n &\text{ proper}, & \deg(f_n) &= n. \end{aligned}$$

$$X = Y \implies |n| = 1 \xrightarrow{6.1.3} f \in \text{Aut}(X), \quad X = A(r, R) \implies f(z) = e^{i\theta} z \quad \text{or} \quad f(z) = e^{i\theta} \frac{rR}{z}.$$

This proves existence and self-map rigidity. $\square$

7. BOUNDARY INTERACTION AND DEGENERATION

For $A := A(r, R)$, put $m := \text{mod}(A)$, $W := \log(R/r) = 2\pi m$, and $t := \log(|z|/r)$; constant boundary data propagates linearly in t/W , whereas nonconstant modes decay exponentially with their distance from the supplying boundary [2, 17].

7.1. Transfer and limiting regimes.

Theorem 7.1.1 (Annular boundary transfer, [2, 17, 18]). *Let $\varphi_r, \varphi_R \in C(\mathbb{R}/(2\pi\mathbb{Z}), \mathbb{R})$. Define their Fourier coefficients and, for $n \neq 0$, the transfer factors by*

$$\widehat{\varphi}_j(n) := \frac{1}{2\pi} \int_0^{2\pi} \varphi_j(\theta) e^{-in\theta} d\theta, \quad P_n^r(t) := \frac{\sinh(|n|(W-t))}{\sinh(|n|W)}, \quad P_n^R(t) := \frac{\sinh(|n|t)}{\sinh(|n|W)}.$$

The unique harmonic $u \in C^2(A) \cap C(\bar{A})$ with boundary values φ_r, φ_R satisfies, for $0 < t < W$,

$$u(re^{t+i\theta}) = \left(1 - \frac{t}{W}\right) \widehat{\varphi}_r(0) + \frac{t}{W} \widehat{\varphi}_R(0) + \sum_{n \neq 0} (P_n^r(t) \widehat{\varphi}_r(n) + P_n^R(t) \widehat{\varphi}_R(n)) e^{in\theta},$$

where the series converges absolutely and locally uniformly. The boundary-component harmonic measures and transfer bounds are

$$\omega_R(re^{t+i\theta}) = \frac{t}{W}, \quad \omega_r(re^{t+i\theta}) = 1 - \frac{t}{W}, \quad \omega_R = u_* \xrightarrow{3.1.4} \mathcal{E}_A(\omega_R) = \text{cap}(A) = m^{-1}.$$

$$0 \leq P_n^r(t) \leq e^{-|n|t}, \quad 0 \leq P_n^R(t) \leq e^{-|n|(W-t)}, \quad P_n^r(W/2) = P_n^R(W/2) = \frac{1}{2 \cosh(|n|W/2)} \leq e^{-\pi|n|m}.$$

Proof. It suffices to solve trigonometric-polynomial data and use Fejér approximation.

$$v(t, \theta) := u(re^{t+i\theta}), \quad \Delta u = 0 \iff \left(\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial \theta^2} \right) v = 0, \quad v(t, \theta) = \sum_{n \in \mathbb{Z}} v_n(t) e^{in\theta}.$$

$$v_0'' = 0 \implies v_0(t) = \left(1 - \frac{t}{W}\right) \widehat{\varphi}_r(0) + \frac{t}{W} \widehat{\varphi}_R(0), \quad v_n'' - n^2 v_n = 0 \implies v_n(t) = P_n^r(t) \widehat{\varphi}_r(n) + P_n^R(t) \widehat{\varphi}_R(n).$$

$$P_n^r(t) = e^{-|n|t} \frac{1 - e^{-2|n|(W-t)}}{1 - e^{-2|n|W}} \leq e^{-|n|t}, \quad P_n^R(t) = e^{-|n|(W-t)} \frac{1 - e^{-2|n|t}}{1 - e^{-2|n|W}} \leq e^{-|n|(W-t)}.$$

$$|\widehat{\varphi}_j(n)| \leq \|\varphi_j\|_\infty, \quad \Phi_N := (\sigma_N \varphi_r, \sigma_N \varphi_R), \quad \Phi := (\varphi_r, \varphi_R), \quad \|\Phi_N - \Phi\|_\infty \longrightarrow 0.$$

$$\Delta u_N = 0, \quad u_N|_{\partial A} = \Phi_N, \quad \|u_N - u_M\|_\infty \leq \|\Phi_N - \Phi_M\|_\infty \implies u_N \longrightarrow u \text{ uniformly.}$$

This proves the Dirichlet formula and its exact transfer bounds. $\square$

Proposition 7.1.2 (Two-boundary majorization). *Let $f \in \text{Hol}(A, \mathbb{C}) \cap C(\bar{A})$, $M_j := \max_{|z|=j} |f(z)| > 0$, and $\rho = re^t$. Then*

$$|f(\rho e^{i\theta})| \leq M_r^{1-t/W} M_R^{t/W}.$$

If $f, g \in \text{Hol}(A, \mathbb{C}) \cap C(\bar{A})$ are nonvanishing on $\bar{A}$ and

$$\varepsilon_j := \max_{|z|=j} |\log |f(z)| - \log |g(z)|| \quad (j \in \{r, R\}),$$

then

$$\left| \log \left| \frac{f}{g} \right| (re^{t+i\theta}) \right| \leq \left(1 - \frac{t}{W}\right) \varepsilon_r + \frac{t}{W} \varepsilon_R.$$

Proof. The harmonic measures in Theorem 7.1.1 are the sharp barriers.

$$\begin{aligned} \omega_r &= 1 - \frac{t}{W}, \quad \omega_R = \frac{t}{W}, \quad L_f := \omega_r \log M_r + \omega_R \log M_R. \\ \log |f| - L_f &\text{ is subharmonic,} \quad \log |f| - L_f \leq 0 \text{ on } \partial A \implies \log |f| \leq L_f. \\ h &:= \log \left| \frac{f}{g} \right|, \quad H := \omega_r \varepsilon_r + \omega_R \varepsilon_R, \quad \Delta h = \Delta H = 0, \quad -H \leq h \leq H \text{ on } \partial A \implies |h| \leq H. \end{aligned}$$

This proves both weighted maximum-principle estimates. $\square$

Theorem 7.1.3 (Long-cylinder degeneration and removability, [7, 18]). *Let $m_k \rightarrow \infty$, $q_k := e^{-2\pi m_k}$, $s_k := \sqrt{q_k}$, and $f_k \in \text{Hol}(A(q_k, 1), \mathbb{D})$, with*

$$f_k(z) = \sum_{n \in \mathbb{Z}} a_n^{(k)} z^n, \quad b_n^{(k)} := s_k^n a_n^{(k)}.$$

Then

$$\begin{aligned} |b_n^{(k)}| &\leq e^{-\pi |n| m_k}, \quad |a_n^{(k)}| \leq 1 \quad (n \geq 0), \quad |a_{-n}^{(k)}| \leq q_k^n \quad (n \geq 1), \\ \sup_{|z|=s_k} |f_k(z) - a_0^{(k)}| &\leq \frac{2s_k}{1-s_k}. \end{aligned}$$

Every such sequence has a subsequence and a function $f_\infty \in \text{Hol}(\mathbb{D}, \mathbb{C})$, $f_\infty(\mathbb{D}) \subset \overline{\mathbb{D}}$, for which

$$f_{k_j} \longrightarrow f_\infty \text{ locally uniformly on } \mathbb{D}^*, \quad a_0^{(k_j)} \longrightarrow f_\infty(0), \quad \sup_{\theta \in \mathbb{R}} |f_{k_j}(s_{k_j} e^{i\theta}) - f_\infty(0)| \longrightarrow 0.$$

Moreover, every bounded $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n \in \text{Hol}(\mathbb{D}^, \mathbb{C})$ satisfies*

$$|a_{-n}| \leq M q^n = M e^{-2\pi n \text{mod}(A(q, 1))} \quad (0 < q < 1, n \geq 1), \quad M := \|f\|_{L^\infty(\mathbb{D}^*)}, \quad a_{-n} = 0,$$

and therefore extends holomorphically across 0.

Proof. Fix $0 < \rho < 1$; centered damping, Montel compactness, and Cauchy's formula control both limits.

$$\begin{aligned} |b_n^{(k)}| &\xrightarrow{4.1.3} s_k^{|n|} = e^{-\pi |n| m_k}, \quad b_n^{(k)} = s_k^n a_n^{(k)} \implies |a_n^{(k)}| \leq 1 \quad (n \geq 0), \quad |a_{-n}^{(k)}| \leq s_k^{2n} = q_k^n \quad (n \geq 1). \\ \sup_{|z|=s_k} |f_k - a_0^{(k)}| &\xrightarrow{4.1.4} \frac{2s_k}{1-s_k}. \end{aligned}$$

$$a_{-n} = \frac{1}{2\pi i} \int_{|z|=q} f(z) z^{n-1} dz, \quad |a_{-n}| \leq M q^n \quad (0 < q < 1), \quad q \downarrow 0 \implies a_{-n} = 0 \quad (n \geq 1).$$

$$0 < \alpha < 1, \quad A(\alpha, 1) \subset A(q_k, 1) \quad (k \gg 1), \quad |f_k| \leq 1 \xrightarrow{\text{Montel/diagonal}} f_{k_j} \xrightarrow{\text{loc.}} f_\infty \text{ on } \mathbb{D}^*, \quad |f_\infty| \leq 1.$$

$$c_j := a_0^{(k_j)} = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{f_{k_j}(z)}{z} dz \longrightarrow f_\infty(0), \quad \sup_{\theta} |f_{k_j}(s_{k_j} e^{i\theta}) - f_\infty(0)| \leq \frac{2s_{k_j}}{1-s_{k_j}} + |c_j - f_\infty(0)| \longrightarrow 0.$$

This proves compactness, core collapse, and removability. $\square$

For the opposite degeneration $m \downarrow 0$, let $s := \sqrt{rR}$ and $F \in \text{Hol}(A, \mathbb{D})$. The intrinsic formulas of § 5 give

$$s \rho_A(s e^{i\theta}) = \frac{1}{2m}, \quad \ell_A(\{|z|=s\}) = \frac{\pi}{m}, \quad \frac{s |F'(s e^{i\theta})|}{1 - |F(s e^{i\theta})|^2} \leq \frac{1}{4m}.$$

The derivative order is attained up to constants by

$$0 < m < \pi^{-1}, \quad N_m := \left\lfloor \frac{1}{\pi m} \right\rfloor, \quad G_m(z) := e^{-\pi m N_m} z^{N_m} \in \text{Hol}(A(e^{-\pi m}, e^{\pi m}), \mathbb{D}).$$

$$m |G'_m(1)| = m N_m e^{-\pi m N_m} \longrightarrow \frac{1}{\pi e}.$$

Thus long cylinders suppress nonconstant modes at the scale $e^{-\pi m}$, whereas short cylinders have intrinsic scale m^{-1} .

8. QUASICONFORMAL STABILITY AND OUTLOOK

Quasiconformal deformation preserves annular invariants up to maximal dilatation, and $\frac{1}{2} \log \text{mod}(A)$ is the exact Teichmüller coordinate [2, 13, 10].

8.1. Sharp modulus distortion.

Definition 8.1.1 (Quasiconformal map, [13, 4]). Let $\Omega, \Omega' \subset \mathbb{C}$ and $K \geq 1$. An orientation-preserving homeomorphism $f : \Omega \rightarrow \Omega'$ with $f \in W_{\text{loc}}^{1,2}(\Omega, \mathbb{C})$ is K -quasiconformal if

$$\left| \left(\frac{\partial}{\partial \bar{z}} f \right) (z) \right| \leq \frac{K-1}{K+1} \left| \left(\frac{\partial}{\partial z} f \right) (z) \right| \quad \text{for almost every } z \in \Omega.$$

Equivalently, with $\|Df(z)\|$ the operator norm and $J_f(z) := \det(Df(z))$,

$$\|Df(z)\|^2 \leq K J_f(z) \quad \text{a.e.}, \quad K(f) := \inf\{K \geq 1 : f \text{ is } K\text{-quasiconformal}\}.$$

Theorem 8.1.2 (Extremal-length distortion, [2, 13]). Let $f : \Omega \rightarrow \Omega'$ be K -quasiconformal and let Γ be a curve family in Ω . Then

$$\frac{1}{K} \text{Ext}_{\Omega}(\Gamma) \leq \text{Ext}_{\Omega'}(f(\Gamma)) \leq K \text{Ext}_{\Omega}(\Gamma).$$

Proof. Fix $\rho' \geq 0$ with $0 < \text{Area}_{\rho'}(\Omega') < \infty$; the ACL chain rule and a Fuglede correction control a zero-2-modulus exceptional family.

$$\ell_{\rho'}(f \circ \gamma) \leq \int_{\gamma} \rho'(f(z)) \|Df(z)\| |dz| \quad (\gamma \notin \Gamma_0), \quad h \in L^2(\Omega), \quad h \geq 0, \quad \int_{\gamma} h |dz| = \infty \quad (\gamma \in \Gamma_0).$$

$$\rho_{\varepsilon}(z) := \rho'(f(z)) \|Df(z)\| + \varepsilon h(z), \quad L_{\rho_{\varepsilon}}(\Gamma) \geq L_{\rho'}(f(\Gamma)).$$

$$\text{Area}_{\rho_{\varepsilon}}(\Omega) \longrightarrow \int_{\Omega} (\rho' \circ f)^2 \|Df\|^2 dA \leq K \int_{\Omega} (\rho' \circ f)^2 J_f dA = K \text{Area}_{\rho'}(\Omega').$$

$$\text{Ext}_{\Omega}(\Gamma) \geq \lim_{\varepsilon \downarrow 0} \frac{L_{\rho_{\varepsilon}}(\Gamma)^2}{\text{Area}_{\rho_{\varepsilon}}(\Omega)} \geq \frac{1}{K} \frac{L_{\rho'}(f(\Gamma))^2}{\text{Area}_{\rho'}(\Omega')} \implies \text{Ext}_{\Omega'}(f(\Gamma)) \leq K \text{Ext}_{\Omega}(\Gamma).$$

$$K(f^{-1}) = K(f) \leq K \implies \text{Ext}_{\Omega}(\Gamma) \leq K \text{Ext}_{\Omega'}(f(\Gamma)) \implies \text{Ext}_{\Omega'}(f(\Gamma)) \geq K^{-1} \text{Ext}_{\Omega}(\Gamma).$$

This proves both distortion inequalities. $\square$

Corollary 8.1.3 (Sharp annular distortion). Let $f : A \rightarrow A'$ be K -quasiconformal, $m := \text{mod}(A)$, and $m' := \text{mod}(A')$. Then

$$\frac{m}{K} \leq m' \leq Km, \quad \frac{\text{cap}(A)}{K} \leq \text{cap}(A') \leq K \text{cap}(A).$$

Among quasiconformal homeomorphisms $g : A \rightarrow A'$, the optimum and Teichmüller distance are

$$K_{\min}(A, A') := \inf_g K(g) = \max \left\{ \frac{m'}{m}, \frac{m}{m'} \right\}, \quad d_T(A, A') := \frac{1}{2} \log K_{\min}(A, A') = \frac{1}{2} \left| \log \frac{m'}{m} \right|.$$

Proof. Conformally identify A, A' with the centered round annuli below; Theorems 8.1.2 and 3.1.4, applied to essential loops modulo zero 2-modulus, give

$$\frac{1}{Km} \leq \frac{1}{m'} \leq \frac{K}{m} \iff \frac{m}{K} \leq m' \leq Km.$$

$$\lambda := \frac{m'}{m}, \quad F_{\lambda}(z) := |z|^{\lambda-1} z : A(e^{-\pi m}, e^{\pi m}) \longrightarrow A(e^{-\pi m'}, e^{\pi m'}).$$

$$\tilde{F}_{\lambda}(x + iy) := \lambda x + iy, \quad D\tilde{F}_{\lambda} = \text{diag}(\lambda, 1) \implies K(F_{\lambda}) = \max\{\lambda, \lambda^{-1}\}.$$

$$K(g) \geq \max\{\lambda, \lambda^{-1}\}, \quad K(F_\lambda) = \max\{\lambda, \lambda^{-1}\} \implies K_{\min}(A, A') = K(F_\lambda), \quad d_T(A, A') = \frac{1}{2} |\log \lambda|.$$

The capacity identities give the remaining bounds, while the stretch proves sharpness and the distance formula. $\square$

Remark 8.1.4 (Further directions). Natural extensions are the exact Carathéodory density and its extremizers,

$$c_A(z) := \sup_{g \in \text{Hol}(A, \mathbb{D})} \frac{2|g'(z)|}{1 - |g(z)|^2} \leq \rho_A(z).$$

9. PROOF OF THEOREM A

Theorem 9.0.1 (Modulus control). *Let $A = A(r, R)$, $A' = A(r', R')$, $m = \text{mod}(A)$, $W = 2\pi m$, and $q = r/R$. Then*

(1) *For the connecting and separating curve families,*

$$\text{Ext}_A(\Gamma_{\text{rad}}) = m, \quad \text{Ext}_A(\Gamma_{\text{sep}}) = \text{cap}(A) = m^{-1}.$$

Proof.

$$A = A(r, R), \quad m = \text{mod}(A) \xrightarrow{3.1.4} \text{Ext}_A(\Gamma_{\text{rad}}) = m, \quad \text{Ext}_A(\Gamma_{\text{sep}}) = \text{cap}(A) = m^{-1}.$$

$\square$

(2) *If $f \in \text{Hol}(A(q, 1), \mathbb{C})$, $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$, and $\sup_{A(q, 1)} |f| \leq M$, then*

$$|a_n| \leq M \quad (n \geq 0), \quad |a_{-n}| \leq M q^n \quad (n \geq 1),$$

$$\sup_{|z|=\sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1 - \sqrt{q}}.$$

Proof. Put $s := \sqrt{q} = e^{-W/2}$ and $b_k := s^k a_k$. Then

$$b_k = s^k a_k \xrightarrow{4.1.2, 4.1.3} |b_k| \leq M s^{|k|}.$$

Hence, for $n \geq 0$ and $j \geq 1$,

$$|a_n| = s^{-n} |b_n| \leq M, \quad |a_{-j}| = s^j |b_{-j}| \leq M s^{2j} = M q^j.$$

Moreover,

$$e^{-\pi m} = e^{-W/2} = \sqrt{q} \xrightarrow{4.1.4} \sup_{|z|=\sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1 - \sqrt{q}}.$$

$\square$

(3) *For $C_* := \{|z| = \sqrt{rR}\}$,*

$$\rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log\left(\frac{|z|}{r}\right)\right), \quad \ell_A(C_*) = \frac{\pi}{m}.$$

Moreover, if $f \in \text{Hol}(A, \mathbb{D})$ and $\rho_{\mathbb{D}}(\xi) := 2(1 - |\xi|^2)^{-1}$, then

$$\rho_{\mathbb{D}}(f(z)) |f'(z)| \leq \rho_A(z), \quad d_{\mathbb{D}}(f(z), f(\zeta)) \leq d_A(z, \zeta).$$

Proof.

$$W = 2\pi m, \quad C_* = \{|z| = \sqrt{rR}\} \xrightarrow{5.1.2, 5.1.3} \rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log\left(\frac{|z|}{r}\right)\right), \quad \ell_A(C_*) = \frac{\pi}{m}.$$

$$f \in \text{Hol}(A, \mathbb{D}) \xrightarrow{5.1.4} \rho_{\mathbb{D}}(f(z)) |f'(z)| \leq \rho_A(z), \quad d_{\mathbb{D}}(f(z), f(\zeta)) \leq d_A(z, \zeta).$$

$\square$

(4) If $g \in \text{Hol}(A, A')$ is proper of signed degree $n \in \mathbb{Z} \setminus \{0\}$, then, in round coordinates,

$$g(z) = c \cdot z^n, \quad c \in \mathbb{C} \setminus \{0\}, \quad \text{mod}(A') = |n| \cdot \text{mod}(A).$$

Proof.

$$g \in \text{Hol}(A, A') \text{ proper}, \quad n = \deg(g) \neq 0 \xrightarrow{6.1.3} g(z) = c \cdot z^n, \quad c \neq 0, \quad \text{mod}(A') = |n| \cdot \text{mod}(A).$$

□

(5) If $h : A \rightarrow A'$ is a K -quasiconformal homeomorphism, then

$$K^{-1} \cdot \text{mod}(A) \leq \text{mod}(A') \leq K \cdot \text{mod}(A).$$

Proof.

$$h : A \rightarrow A' \text{ is } K\text{-quasiconformal} \xrightarrow{8.1.3} K^{-1} \cdot \text{mod}(A) \leq \text{mod}(A') \leq K \cdot \text{mod}(A).$$

□

Part 2. Appendix

APPENDIX A. FOUNDATIONAL TOOLS

A.1. Cauchy bounds, coverings, and scales.

Lemma A.1.1 (Cauchy bounds, [7, 18]). *Let $U \subset \mathbb{C}$ be a domain, $\overline{D(z_0, d)} \subset U$, and $f \in \text{Hol}(U, \mathbb{C})$. Then*

$$|f^{(m)}(z_0)| \leq \frac{m!}{d^m} \max_{|z-z_0|=d} |f(z)| \quad (m \in \mathbb{Z}_{\geq 0}).$$

If $g(z) = \sum_{n \in \mathbb{Z}} a_n z^n \in \text{Hol}(A(r, R), \mathbb{C})$, then

$$|a_n| \leq \rho^{-n} \max_{|z|=\rho} |g(z)| \quad (r < \rho < R, n \in \mathbb{Z}).$$

Proof. Cauchy's integral formulas give both estimates directly.

$$\begin{aligned} f^{(m)}(z_0) &= \frac{m!}{2\pi i} \int_{|z-z_0|=d} \frac{f(z)}{(z-z_0)^{m+1}} dz \implies |f^{(m)}(z_0)| \leq \frac{m!}{d^m} \max_{|z-z_0|=d} |f(z)|. \\ a_n &= \frac{1}{2\pi i} \int_{|z|=\rho} \frac{g(z)}{z^{n+1}} dz \implies |a_n| \leq \rho^{-n} \max_{|z|=\rho} |g(z)|. \end{aligned}$$

This proves both bounds. $\square$

Lemma A.1.2 (Periodic extraction, [7, 9]). *Let $a < b$ be real and $S_{a,b} := \{w \in \mathbb{C} : a < \Re(w) < b\}$. Then*

$$e^{w_1} = e^{w_2} \iff w_2 - w_1 \in 2\pi i \mathbb{Z},$$

and every $2\pi i$ -periodic $F \in \text{Hol}(S_{a,b}, \mathbb{C})$ has the locally absolutely and uniformly convergent expansion

$$F(w) = \sum_{n \in \mathbb{Z}} b_n e^{nw}, \quad b_n(x) := \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny} dy \equiv b_n \quad (a < x < b).$$

Moreover, if Ω is connected and $h \in \text{Hol}(\Omega, \mathbb{C})$ satisfies $h(\Omega) \subset \mathbb{Z}$, then h is constant.

Proof. The fiber relation, discreteness, descent, and Laurent expansion give the claims.

$$\begin{aligned} e^{w_2-w_1} &= 1, \quad w_2 - w_1 = u + iv \iff e^u = 1, \quad e^{iv} = 1 \iff u = 0, \quad v \in 2\pi \mathbb{Z}. \\ \Omega \text{ connected, } h \text{ continuous, } h(\Omega) &\subset \mathbb{Z} \implies h(\Omega) \text{ connected} \implies h(\Omega) = \{k\}. \\ f(e^w) &:= F(w), \quad e^{w_1} = e^{w_2} \implies F(w_1) = F(w_2), \quad F = f \circ \exp, \quad f \in \text{Hol}(A(e^a, e^b), \mathbb{C}). \\ z = e^{x+iy}, \quad dz &= iz dy \implies \frac{1}{2\pi i} \int_{|z|=e^x} \frac{f(z)}{z^{n+1}} dz = \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny} dy = b_n(x). \\ f(z) &= \sum_{n \in \mathbb{Z}} b_n z^n, \quad b_n(x) \equiv b_n \implies F(w) = f(e^w) = \sum_{n \in \mathbb{Z}} b_n e^{nw}. \end{aligned}$$

The convergence is locally absolute and uniform, proving the fiber, constancy, and Fourier assertions. $\square$

Proposition A.1.3 (Equal-modulus shells). *Let $A := A(r, R)$, $W := \log(R/r) = 2\pi \text{mod}(A)$, and $N \in \mathbb{Z}_{\geq 1}$. The radii*

$$r_j := r e^{jW/N} \quad (0 \leq j \leq N)$$

cut A into shells $A_j := A(r_j, r_{j+1})$, $0 \leq j < N$, satisfying

$$\text{mod}(A_j) = \frac{\text{mod}(A)}{N}.$$

Under $p_r : S_{0,W} \rightarrow A$, $p_r(w) := r e^w$, these shells are equal-width cells. With $\pi_{12}(X_1, X_2, X_3) := X_1 + iX_2$, the cover has the helicoidal realization

$$\Psi_r(x+iy) := (r e^x \cos y, r e^x \sin y, y), \quad \pi_{12} \circ \Psi_r = p_r, \quad \Psi_r(w+2\pi i) = \Psi_r(w) + (0, 0, 2\pi).$$

Proof. Both conclusions follow by taking logarithmic radius.

$$\frac{r_{j+1}}{r_j} = e^{W/N} \implies \text{mod}(A_j) = \frac{1}{2\pi} \log \frac{r_{j+1}}{r_j} = \frac{W}{2\pi N} = \frac{\text{mod}(A)}{N}.$$

$$\log \frac{|p_r(x+iy)|}{r} = x, \quad \frac{jW}{N} < x < \frac{(j+1)W}{N} \iff r_j < |p_r(x+iy)| < r_{j+1}.$$

$$\pi_{12}(\Psi_r(x+iy)) = re^x(\cos y + i \sin y) = p_r(x+iy), \quad \Psi_r(x+i(y+2\pi)) = \Psi_r(x+iy) + (0, 0, 2\pi).$$

This proves the shell and helicoid formulas. $\square$

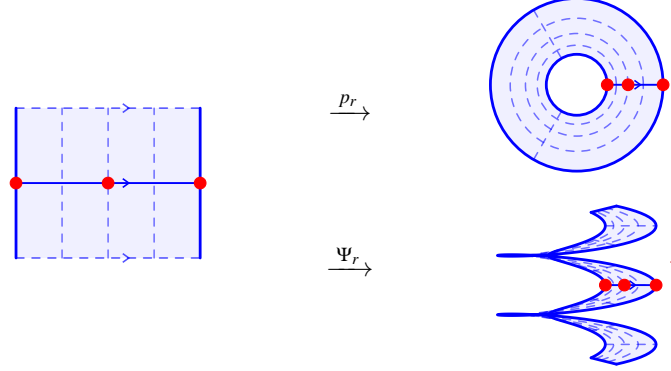


FIGURE 5. Equal-width strip cells map to equal-modulus shells under p_r and to helicoidal bands under Ψ_r ; the red arrow records $y \mapsto y + 2\pi$.

Proposition A.1.4 (Spherical compactification). *Inverse stereographic projection is*

$$\Sigma(z) := \frac{(2\Re(z), 2\Im(z), |z|^2 - 1)}{1 + |z|^2} : \widehat{\mathbb{C}} \longrightarrow \mathbb{S}^2, \quad \Sigma(\infty) := (0, 0, 1).$$

For $h_\rho := (\rho^2 - 1)/(\rho^2 + 1)$,

$$\Sigma(A(r, R)) = \{X \in \mathbb{S}^2 : h_r < X_3 < h_R\},$$

and

$$\text{mod}(A(r, R)) = \frac{1}{4\pi} \log \frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)}.$$

In particular, $\Sigma(A(q, 1)) = \{X \in \mathbb{S}^2 : h_q < X_3 < 0\}$, and $h_q \downarrow -1$ as $q \downarrow 0$.

Proof. The third coordinate records radius monotonically.

$$X_1^2 + X_2^2 = \frac{4\rho^2}{(1+\rho^2)^2}, \quad X_3 = h_\rho, \quad X_1^2 + X_2^2 + X_3^2 = 1, \quad h'_\rho = \frac{4\rho}{(1+\rho^2)^2} > 0.$$

$$h_\rho = \frac{\rho^2 - 1}{\rho^2 + 1} \iff \rho^2 = \frac{1+h_\rho}{1-h_\rho}.$$

$$\frac{R}{r} = \left(\frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)} \right)^{1/2} \implies \text{mod}(A(r, R)) = \frac{1}{4\pi} \log \frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)}.$$

This proves the band and modulus formulas. $\square$

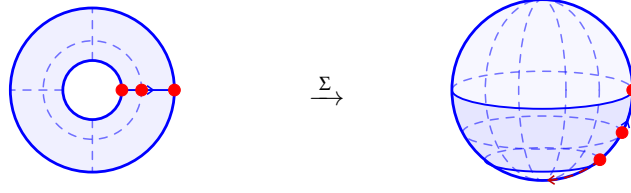


FIGURE 6. Inverse stereographic projection sends $|z| = \rho$ to $X_3 = h_\rho$, so $A(q, 1)$ becomes $h_q < X_3 < 0$, with $h_q \downarrow -1$ as $q \downarrow 0$.

Proposition A.1.5 (Certified test family). *Let $A := A(q, 1)$, $W := \log(1/q) = 2\pi \bmod(A)$, $s := \sqrt{q}$, $N \in \mathbb{Z}_{\geq 0}$, $0 < \eta < 1$, and $\xi_{-N}, \dots, \xi_N \in \mathbb{C}$ not all zero. Put*

$$C := \sum_{|n| \leq N} |\xi_n|, \quad b_n := \frac{\eta e^{-|n|W/2} \xi_n}{C}, \quad f_N(z) := \sum_{|n| \leq N} b_n \left(\frac{z}{s}\right)^n.$$

Then $f_N \in \text{Hol}(A, \mathbb{D})$, $\|f_N\|_{L^\infty(A)} \leq \eta$, and $|b_n| \leq \eta e^{-|n|W/2}$. With

$$\Delta_A(z) := \frac{W}{2} - \left| \log \frac{|z|}{s} \right|, \quad \kappa_A(z) := \frac{1}{|z| \Delta_A(z)}, \quad u := \frac{\pi \Delta_A(z)}{W},$$

the exact diagnostics are

$$\frac{\rho_A(z)}{\kappa_A(z)} = \frac{u}{\sin u} \in \left[1, \frac{\pi}{2}\right], \quad Q_N(z) := \frac{2|f'_N(z)|}{(1 - |f_N(z)|^2)\rho_A(z)} \leq 1.$$

Proof. The centered coordinate certifies boundedness before any sampling.

$$z = se^{x+iy}, \quad |x| < \frac{W}{2} \implies \left| b_n \left(\frac{z}{s}\right)^n \right| \leq \frac{\eta |\xi_n|}{C} e^{-|n|W/2 + nx} \leq \frac{\eta |\xi_n|}{C}.$$

$$|f_N(z)| \leq \frac{\eta}{C} \sum_{|n| \leq N} |\xi_n| = \eta < 1, \quad |b_n| \leq \eta e^{-|n|W/2}.$$

$$\rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi \Delta_A(z)}{W}\right), \quad \frac{\rho_A(z)}{\kappa_A(z)} = \frac{u}{\sin u} \xrightarrow{5.1.4} Q_N(z) \leq 1.$$

This proves all certified bounds. □

For $q = e^{-2\pi m}$, the exact first-mode scale is $\delta_1(m) := e^{-\pi m} = \sqrt{q}$; the rounded benchmarks are

m	.5	1.5	2.5	3.5	4.5
$q = e^{-2\pi m}$	4.321×10^{-2}	8.070×10^{-5}	1.507×10^{-7}	2.814×10^{-10}	5.255×10^{-13}
$\delta_1(m)$	2.079×10^{-1}	8.983×10^{-3}	3.882×10^{-4}	1.678×10^{-5}	7.249×10^{-7}

TABLE 1. The inner radius and first-mode damping factor decay exponentially with modulus.

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